

Universal Weed Detection and Autonomous Laser Removal Robot Using Foundation Models and Multimodal Computer Vision for Multi-Crop Agricultural Systems

Problem Statement

Weed infestation is a major challenge affecting agricultural productivity worldwide, causing significant yield losses and increased production costs across virtually all crop production systems. Existing weed management practices predominantly rely on herbicide applications and manual or mechanical weeding. These approaches suffer from several limitations, including environmental pollution, herbicide resistance, high labor requirements, soil degradation, and increasing operational costs.

Recent advances in agricultural robotics and computer vision have enabled the development of autonomous weed detection and removal systems. However, the vast majority of existing research focuses on crop-specific weed detection models trained for individual crops such as cotton, maize, soybean, sugar beet, rice, or wheat. These models typically perform well only within the environments and crop types on which they were trained.

The fundamental challenge arises from the enormous variability encountered in agricultural environments. Weed species exhibit substantial diversity in:

- Morphology
- Leaf structure
- Growth stages
- Color characteristics
- Canopy architecture
- Seasonal variations

Similarly, crop species differ significantly in:

- Plant geometry
- Plant density
- Row spacing
- Growth patterns
- Environmental conditions

Consequently, a weed detection model trained in one crop often experiences substantial performance degradation when deployed in another crop. Current approaches require extensive collection and annotation of crop-specific datasets, making large-scale deployment economically impractical.

Furthermore, emerging invasive weed species and previously unseen weed varieties continuously appear in agricultural ecosystems. Traditional supervised deep learning systems struggle to identify these unseen classes because they can only recognize categories present in their training datasets.

Although recent developments in foundation models, vision transformers, self-supervised learning, multimodal sensing, and vision-language models have demonstrated remarkable generalization capabilities in computer vision, their application to universal weed detection remains largely unexplored. There is currently no robust field-deployable system capable of detecting, classifying, and localizing diverse weed species across multiple crop types under varying environmental conditions while simultaneously supporting autonomous robotic intervention.

Therefore, there is a critical need to develop a universal weed detection framework capable of generalizing across crops, weed species, growth stages, geographical locations, and environmental

conditions. Such a system would represent a significant advancement toward scalable precision agriculture by enabling a single autonomous robotic platform to perform weed management across diverse farming systems without requiring crop-specific retraining.

Research Gap

Current weed detection systems suffer from several limitations:

Existing Research Focuses On:

- Cotton weed detection
- Wheat weed detection
- Soybean weed detection
- Sugar beet weed detection
- Rice weed detection

Current Limitations

1. Crop-specific training.
2. Poor transferability across crops.
3. Large annotation requirements.
4. Inability to recognize unseen weed species.
5. Limited robustness under varying field conditions.
6. Lack of multimodal sensing integration.
7. Limited use of foundation models.
8. Few autonomous intervention systems.

Missing Capability

A single AI model capable of:

Any Crop + Any Weed Species + Any Growth Stage
under real-world field conditions.

Novel Research Question

Can a foundation-model-based multimodal perception system learn generalized crop-weed representations that enable zero-shot or few-shot weed detection across diverse crop production systems without crop-specific retraining?

Aim

To develop a universal AI-powered agricultural robotic system capable of detecting and removing weed species across multiple crop types using foundation models, multimodal sensing, and autonomous laser actuation.

Objectives

Objective 1:

Develop a large-scale multi-crop weed dataset including:

- Cotton
- Wheat
- Rice
- Maize
- Soybean
- Sugarcane
- Vegetable crops

Objective 2

Develop a universal weed detection model using:

- Vision Transformers (ViTs)
- DINOv2
- Segment Anything Model (SAM)
- Florence
- Agricultural Vision-Language Models

Objective 3

Integrate multimodal sensing:

- RGB
- Multispectral
- Thermal
- Depth sensors

Objective 4

Develop zero-shot weed detection capability for unseen weed species.

Objective 5

Develop autonomous robotic navigation in multiple crop environments.

Objective 6

Integrate precision laser weed removal.

Objective 7

Evaluate generalization performance across:

- Multiple crops
- Multiple seasons
- Multiple locations

